

ssh

SSH = Secure Shell

هو بروتوكول بيخلك:

- تدخل على جهاز ثاني عن بُعد
- تنفذ أوامر عليه
- بشكل آمن ومشفر

يعني بدل ما تبقى قاعد قدام السيرفر فعليًا، بتتحكم فيه من جهازك

مع SSH

- الاتصال مشفر
- البيانات محمية
- مصادقة قوية

عشان كذا SSH هو الأساس في إدارة السيرفرات

بيتم التواصل بين ال Server + Sys Admin عن طريق port22/TCP

Port / عباره عن رقم بيحدد الخدمة اللي جوا السيرفر
بمعنى أي سيرفر ليه ip و ال port هو عنوان ال serves
بيكون مكون من 0 الي 65535

Port Num	Category
0→1023	Sys Ports

[80 , 443 , 21 , 25]

[http,https,ftp,smtp]

Port Num	Category
1024→49151	User Ports

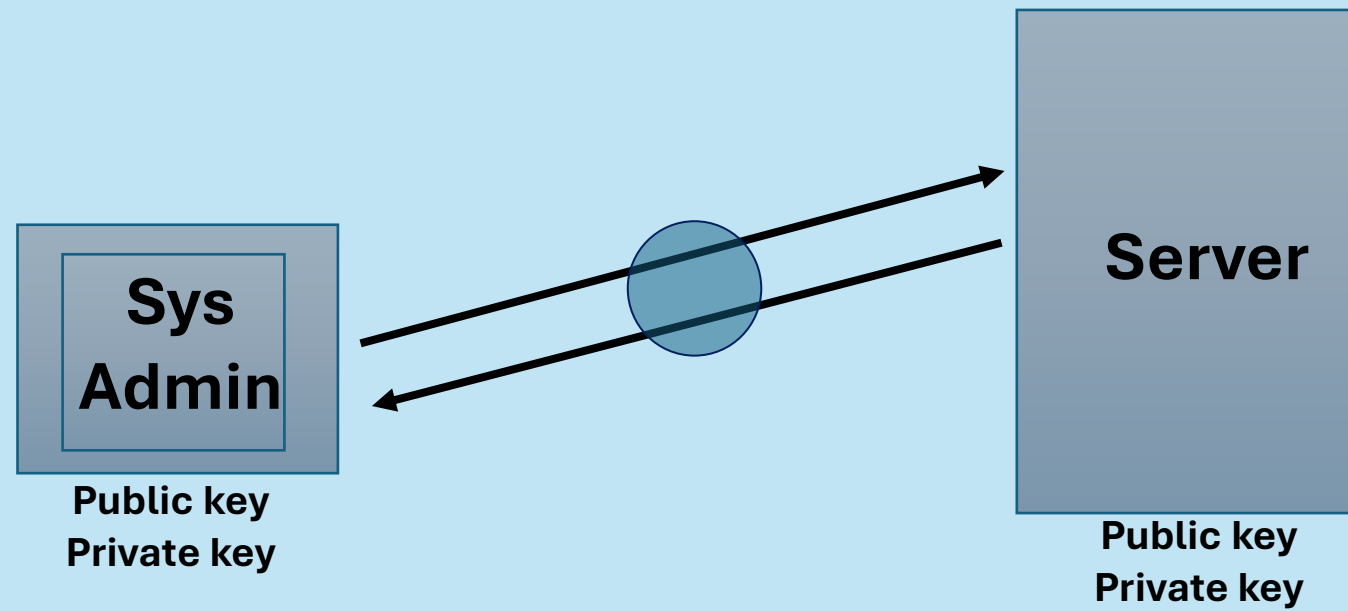
SQL / Oracle

Port Num	Category
49152→65535	Private Ports

ال Users

في عندنا نوعين ال Ports

- TCP / بيستخدم لنقل الداتا بشكل دقيق جدا جدا ولازم يأكد الاستلام الصحيح
- UDP / بيستخدم لنقل الداتا ولكن بشكل غير دقيق



أي جهازين **Server + Computer** يبقى عندهم **Pub key + Priv key** والعلاقة ما بينهم **Pair**

طريقة التواصل الـ **Sys Admin** عايز يتصل بـ **server**

1. **هحمل Ubuntu Server** هصطبه على الـ **VM Ware**

2. **عشان اعمل if config لازم احمل Network tools = tool-net**

3. **هعمل تحديث لقاعدة البيانات**

```
yousef@yousef:~$ sudo apt update
Hit:1 http://eg.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://eg.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,391 kB]
Get:5 http://eg.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://eg.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,684 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [225 kB]
Get:8 http://eg.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [311 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,391 kB]
```

4. **هبدأ اعمل عملية تحميل وتثبيت الـ net-tools**

```
yousef@yousef:~$ sudo apt install net-tools -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
net-tools is already the newest version (2.10-0.1ubuntu4.4).
0 upgraded, 0 newly installed, 0 to remove and 80 not upgraded.
```

كدا احنا جهزنا الـ **machine**

5. **هنستخدم امر ifconfig** عشان نظهر الـ **ip** بتاع الـ **server**

6. **عشان اعمل keys** ليا ومن خلاله هقدر اتواصل مع الـ **servers** واقدر اعمل key لكل **server** على حسب شغلنا

```
[yousef@localhost ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/yousef/.ssh/id_rsa):
```

بيقولك دا **Location** بتاع الـ **private key** → **home/yousef/.ssh/id_rsa/**

وال **public key** بيتعمل تلقائيا معاه → **home/yousef/.ssh/id_rsa.pub/**

هيتطلب منك تدخل password للـ key لو نسيته ساعتها مش هيكون ليه لازمه

```
[yousef@localhost ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/yousef/.ssh/id_rsa):
Created directory '/home/yousef/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/yousef/.ssh/id_rsa
Your public key has been saved in /home/yousef/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:QdllTUKz8V+t/Tjwv0pTNf3ueYvC+HAAHoYRlE/AX+U yousef@localhost.localdomain
The key's randomart image is:
+---[RSA 3072]---+
|      o=+ .o.o0+*o |
|      ooo..o.  o+o |
|      .++o  E   .* |
|      ooo.    o* |
|      .S.    . .+ |
|      .  o.o. |
|      .o.  o+ o |
|      .oo.  o+o |
|      ...o.+* |
+-----[SHA256]-----+
```

انت مخترتش طريقه التشفير اللي هتكون ما بينك ومبين الـ server
اللي هيكون متوصل بالـ key دا .. هو تلقائي اختار ليك الـ rsa

لو عايز اختار نوع التشفير هتستخدم امر وهتضغط بعديها tab مرتين ورا بعض

```
[yousef@localhost ~]$ ssh-keygen -t
option requires an argument -- t
usage: ssh-keygen [-q] [-a rounds] [-b bits] [-C comment] [-f output_keyfile]
                  [-m format] [-N new_passphrase] [-O option]
                  [-t dsa | ecdsa | ecdsa-sk | ed25519 | ed25519-sk | rsa]
                  [-w provider] [-Z cipher]
ssh-keygen -p [-a rounds] [-f keyfile] [-m format] [-N new_passphrase]
            [-P old_passphrase] [-Z cipher]
ssh-keygen -i [-f input_keyfile] [-m key_format]
ssh-keygen -e [-f input_keyfile] [-m key_format]
ssh-keygen -y [-f input_keyfile]
ssh-keygen -c [-a rounds] [-C comment] [-f keyfile] [-P passphrase]
ssh-keygen -l [-v] [-E fingerprint_hash] [-f input_keyfile]
ssh-keygen -B [-f input_keyfile]
ssh-keygen -D pkcs11
ssh-keygen -F hostname [-lv] [-f known_hosts_file]
ssh-keygen -H [-f known_hosts_file]
ssh-keygen -K [-a rounds] [-w provider]
ssh-keygen -R hostname [-f known_hosts_file]
ssh-keygen -r hostname [-g] [-f input_keyfile]
ssh-keygen -M generate [-O option] output file
```

ممکن يواجهك مشكله هيقولك لازم تكتب نوع التشفير بس انت ممكن تعرضهم كلهم وتختار

ميزة الـ shell completion في بعض الأنظمة فقط جرب عندك الاول لو مش تمام هتحملها

```
[yousef@localhost ~]$ sudo yum install bash-completion
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "subscription-man
ager" to register.

Last metadata expiration check: 0:28:04 ago on Wed 24 Dec 2025 02:11:01 AM EET.
Dependencies resolved.
=====
Package                        Architecture Version      Repository    Size
=====
Installing:
bash-completion                noarch       1:2.11-5.el9 Repo2        459 k
Transaction Summary
=====
Install 1 Package
```



```
[yousef@localhost ~]$ ssh-keygen -t dsa / ecdsa / ecdsa-sk / ed25519 / ed25519-sk / rsa /
```

وبكدا ظهرت كل أنواع التشفير هتختار نوع التشفير هتستخدم الأمر بالشكل دا

```
[yousef@localhost ~]$ ssh-keygen -t dsa
Generating public/private dsa key pair.
Enter file in which to save the key (/home/yousef/.ssh/id_dsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/yousef/.ssh/id_dsa
Your public key has been saved in /home/yousef/.ssh/id_dsa.pub
The key fingerprint is:
SHA256:tT880SNj1ey0rFxi0Yul64Eqd8S55aekQHMqY9/JtMs yousef@localhost.localdomain
The key's randomart image is:
+---[DSA 1024]---+
|
|      .
|    . . +
|   S.o.o *
|  . ++=.X o
| + o.*+/* *
| ..+.0oX.@.
| oo.Eo*o
|
+-----[SHA256]-----+
```

وتكمل نفس الخطوات بشكل طبيعي وبكدا انت جهزت كل حاجه

وتهعمل نفس الكلام دا على الـ server

```
yousef@yousef:~$ sudo ssh-keygen -t dsa
[sudo] password for yousef:
Generating public/private dsa key pair.
Enter file in which to save the key (/root/.ssh/id_dsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_dsa
Your public key has been saved in /root/.ssh/id_dsa.pub
The key fingerprint is:
SHA256:GCFVxIxXwKmilwGIwqCJuEGtXonn6vK1zaKMoRLwdwg root@yousef
The key's randomart image is:
+---[DSA 1024]---+
|=o. ..0B++.
|0o.. ...*
|B o.. .o
|.=E+o .o
|+.+o =. S
|..0.= .
|...0..
|+= ..+
|*0+...o
+-----[SHA256]-----+
```

عشان تقدر تعمل connect محتاج 4 حاجات

1. Ip

2. Port num بتاع الـ ssh

3. الـ user name اللي هعمل من خلاله Login

4. الـ Password بتاع الـ User

عشان اعرف الـ Port Num أي

```
[yousef@localhost ~]$ sudo cat /etc/ssh/sshd_config | grep Port
[sudo] password for yousef:
#Port 22
```

هتكتب امر

```
[yousef@localhost ~]$ ssh yousef@192.168.190.131 -p22
The authenticity of host '192.168.190.131 (192.168.190.131)' can't be established.
ED25519 key fingerprint is SHA256:S8b9lWCx25zy7iWAtPMj4Ztbch6Qx5905UbZXytD8bE.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.190.131' (ED25519) to the list of known hosts.
yousef@192.168.190.131's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Wed Dec 24 01:10:00 AM UTC 2025

System load:  0.0      Processes:            215
Usage of /:   60.9% of 9.75GB   Users logged in:     1
Memory usage: 16%      IPv4 address for ens33: 192.168.190.131
Swap usage:   0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

59 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sun Nov  9 14:45:24 2025 from 192.168.190.129
yousef@yousef:~$
```

Yousef: هو اسم الـ user على الـ server

192.168.190.131@ : الـ ip بتاع الـ server

pp- : الـ port num

بكدا انت جوا السيرفر تقدر تعمل اللي انت عايزه


```
yousef@yousef:~$ sudo mkdir /home/ALnaqib
yousef@yousef:~$ ls /home
ALnaqib  yousef
```

هتدخل من الـ server ونشوف

```
yousef@yousef:/home$ ls
ALnaqib  yousef
```

لو عايز اشوف كل الـ Connections

```
yousef@yousef:~$ sudo netstat -antup
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      646/systemd-resolve
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      646/systemd-resolve
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      1/init
tcp        0    196 192.168.190.131:22       192.168.190.137:44900   ESTABLISHED 1846/sshd: yousef [
tcp6       0      0 :::22                   :::*                   LISTEN      1/init
udp        0      0 127.0.0.54:53           0.0.0.0:*               646/systemd-resolve
udp        0      0 127.0.0.53:53           0.0.0.0:*               646/systemd-resolve
udp        0      0 192.168.190.131:68      0.0.0.0:*               627/systemd-network
```

Network Statistics : Netstat

a : هاتلي كل ال connections

n : هاتلي بنظام ال new mark بالأرقام

t : كل أنواع ال Connection = tcp

u : كل أنواع ال connection = udp

p : ال ip بتاع ال session